Ex 1 pg 321 Sketch of solution.

a) $\Delta = 4 - 27 = -23$ not a square in $\mathbb{Q}$ $\Rightarrow G = S_3$

b) $\Delta = -27 \cdot 10^2$ not a square in $\mathbb{Q}$ $\Rightarrow G = S_3$ c) Same as b

d) $\Delta = -27 \cdot 10^2 = (3\sqrt{-3} \cdot 10)^2 \Rightarrow G = \mathbb{Z}_3$ e) $\Delta = (\sqrt{-23})^2 \Rightarrow G = \mathbb{Z}_3$

f) $\mathbb{K} = \mathbb{Q}(\sqrt[4]{5}, i)$ $[\mathbb{K}, \mathbb{Q}] = 8$ check $G(\mathbb{K}, \mathbb{Q}) \cong D_4 = \langle \sigma, \tau \rangle$ $\begin{cases} \tau(z) = \bar{z} \\ \sigma(i) = i \quad \sigma(\sqrt[4]{5}) = i\sqrt[4]{5} \end{cases}$

- $x^4 - 5 = (x^2 - \sqrt{5})(x^2 + \sqrt{5}) \in \mathbb{Q}(\sqrt{5})[x]$ so $\mathbb{K} = E_1 \cdot E_2$ $E_1 = \mathbb{Q}(\sqrt{5})(\sqrt[4]{5})$ $E_2 = \mathbb{Q}(\sqrt{5})(i)$

  $E_i/\mathbb{Q}$ are Galois with group $G(E_i, \mathbb{Q}) = \mathbb{Z}_2 \Rightarrow G(\mathbb{K}, \mathbb{Q}) = \mathbb{Z}_2 \times \mathbb{Z}_2$

- $x^4 - 5 = \mathrm{Irr}(\sqrt[4]{5}, \mathbb{Q}(i), x)$ (As $[\mathbb{Q}(\sqrt[4]{5}, i) : \mathbb{Q}(i)] = 4$!)

  $\mathbb{Q}(i) = \mathbb{K}^{\langle \sigma \rangle} \Rightarrow G(\mathbb{Q}(\sqrt[4]{5}, i), \mathbb{Q}(i)) \cong \langle \sigma \rangle \cong \mathbb{Z}_4$

- $\sigma\tau(\sqrt{-5}) = \sigma\tau(i\sqrt[4]{5}\sqrt[4]{5}) = \sigma(-i\sqrt[4]{5}\sqrt[4]{5}) = -i^3\sqrt[4]{5}\sqrt[4]{5} = i\sqrt{5} = \sqrt{-5} \Rightarrow \mathbb{Q}(\sqrt{-5}) = \mathbb{K}^{\langle \sigma\tau, \sigma^2 \rangle}$

  $\sigma^2(\sqrt{-5}) = \sigma^2(i\sqrt[4]{5}\sqrt[4]{5}) = i \cdot i^2\sqrt[4]{5}\, i^2\sqrt[4]{5} = i\sqrt{5} = \sqrt{-5}$

  $\Rightarrow G(\mathbb{K}, \mathbb{Q}(\sqrt{-5})) \cong \langle \sigma\tau, \sigma^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$

h) $\Delta = -27a^2$ is not a square over $\mathbb{Q}$ $\Rightarrow S_3$

i) $x^4 + 2 = (x^2 + i\sqrt{2})(x^2 - i\sqrt{2}) = (x - \alpha\sqrt[4]{2})(x - \alpha^3\sqrt[4]{2})(x - \alpha^5\sqrt[4]{2})(x - \alpha^7\sqrt[4]{2})$ $\alpha = \cos\frac{2\pi}{8} + i\sin\frac{2\pi}{8}$

$\mathbb{K} = \mathbb{Q}(\alpha^2, \alpha\sqrt[4]{2})$ $[\mathbb{Q}(\alpha\sqrt[4]{2}) : \mathbb{Q}] = 4$ $[\mathbb{K} : \mathbb{Q}(\alpha\sqrt[4]{2})] = 2$ $= \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$

$\Rightarrow |G| = 8$, $G < S_4 \Rightarrow G \cong D_4$ as G is not abelian $\alpha^2 = i$ $\alpha^4 = -1$

as $\mathbb{Q}(\alpha\sqrt[4]{2})/\mathbb{Q}$ is not Galois

$|\mathrm{Gal}(\mathbb{K} : \mathbb{Q}(i))| = 4$ so $\mathbb{Z}_4$ or $\mathbb{Z}_2 \times \mathbb{Z}_2$. Notice that for $g \in \mathrm{Gal}(\mathbb{K}; \mathbb{Q}(i))$ is determined by $g(\alpha\sqrt[4]{2}) = \alpha^j\sqrt[4]{2}$ $j \in 1, 3, 5, 7$ as then $g(\alpha^3\sqrt[4]{2}) = g(i\alpha\sqrt[4]{2}) = i\,g(\alpha\sqrt[4]{2})$ same for $\alpha^5\sqrt[4]{2}$, $\alpha^7\sqrt[4]{2}$ so $\mathrm{Gal} \cong \mathbb{Z}_4$

j) $\mathrm{Gal} \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. Must show $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, $\sqrt{5} \notin \mathbb{Q}(\sqrt{2}, \sqrt{3})$ & $\sqrt{7} \notin \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})(\sqrt{5})$ say $\sqrt{7} = \alpha + \beta\sqrt{5}$ $\alpha, \beta \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$

$7 = \alpha^2 + 5\beta^2$ & $\alpha\beta = 0$ If $\beta = 0 \Rightarrow 7 = \alpha^2$ ie $\sqrt{7} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$ which is an easier problem, contradicts $[\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{7}) : \mathbb{Q}] = 8$

If $\alpha = 0$, then $7 = 5\beta^2$. It is enough to show $[\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{\frac{7}{5}}) : \mathbb{Q}] = 8$

Write $\beta = a + b\sqrt{3}$, then $7 = 5(a^2 + 3b^2 + 2ab\sqrt{3}) \Rightarrow ab = 0$

$b = 0 \Rightarrow 7 = 5a^2$. This is an easier problem, contradicts $[\mathbb{Q}(\sqrt{2}, \sqrt{\frac{7}{5}}) : \mathbb{Q}] = 4$

$a = 0 \Rightarrow 7 = 15b^2$. " " also an " " " $[\mathbb{Q}(\sqrt{2}, \sqrt{\frac{15}{7}}) : \mathbb{Q}] = 4$

k) $\mathbb{Z}_2^n$. It suffices to show that $[\mathbb{Q}(\sqrt{p_1} \dots \sqrt{p_n}):\mathbb{Q}]=2^n$

We show by induction that $[\mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_n/q_n}):\mathbb{Q}]=2^n$ where $p_1 \dots p_n\ q_1 \dots q_n$ are distinct relatively ~~prime~~ (possibly $q_i=1$)

$n=1$ $[\mathbb{Q}(\sqrt{p_1/q_1}):\mathbb{Q}]=2$ since $\alpha=\sqrt{p_1/q_1}$ $\alpha^2=p_1/q_1$ so $irr(\alpha,\mathbb{Q},x)=x^2-p_1/q_1$

Note $q_1x^2-p_1$ is irred by Eisenstein since p_1 is a prime with q_1.

Suppose $K=\mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_n/q_n})=\mathbb{Q}(\sqrt{p_1/q_1} \dots \sqrt{p_{n-1}/q_{n-1}})(\sqrt{p_n/q_n})$ & $[K,\mathbb{Q}]=2^{n-1}$

then write $\sqrt{p_n/q_n}=\left(\alpha+\beta\sqrt{p_{n-1}/q_{n-1}}\right)$ $\alpha,\beta\in\mathbb{Q}\left(\sqrt{p_1/q_1} \dots \sqrt{\frac{p_{n-2}}{q_{n-2}}}\right)$. Then

$\frac{p_n}{q_n}=\alpha^2+\beta^2\frac{p_{n-1}}{q_{n-1}}+2\alpha\beta\sqrt{\frac{p_{n-1}}{q_{n-1}}} \Rightarrow \alpha\beta=0$ If $\beta=0 \Rightarrow \frac{p_n}{q_n}=\alpha^2 \Rightarrow \sqrt{\frac{p_n}{q_n}}\in\mathbb{Q}\left(\sqrt{\frac{p_1}{q_1}} \dots \sqrt{\frac{p_{n-2}}{q_{n-2}}}\right)$

which is impossible by ind. If $\alpha=0 \Rightarrow \frac{p_n}{q_n}=\beta^2\frac{p_{n-1}}{q_{n-1}} \Rightarrow$

$\Rightarrow \frac{p_n q_{n-1}}{q_n p_{n-1}}=\beta^2 \Rightarrow \sqrt{\frac{p_n q_{n-1}}{q_n p_{n-1}}}\in\mathbb{Q}\left(\sqrt{\frac{p_1}{q_1}} \dots \sqrt{\frac{p_{n-2}}{q_{n-2}}}\right)$ which is impossible by ind.

l) $(x^3-2)(x^3-3)(x^2-2)$ has splitting field $\mathbb{Q}(\sqrt{2},\sqrt[3]{2},\sqrt[3]{3},\alpha)$ with $\alpha=-\frac{1}{2}+i\frac{\sqrt{3}}{2}$, $\alpha^3=1$

$\mathbb{Q}(\sqrt[3]{2},\sqrt[3]{3},\alpha)$, $\mathbb{Q}(\sqrt{2})$ are Galois over $\mathbb{Q}$

$\mathbb{Q}(\sqrt[3]{2},\alpha)$, $\mathbb{Q}(\sqrt[3]{3},\alpha)$ are Galois over $\mathbb{Q}$ with groups S_3 so $Gal(x^3-2)(x^3-3) \subset S_3\times S_3$

Let $\sigma_1=(1,2,3)$ $\sigma_2=(4,5,6)$ $\tau=(2,3)(5,6)$, then $Gal(\mathbb{Q}(\sqrt[3]{2},\sqrt[3]{3},\alpha)\ \mathbb{Q})\cong\langle\sigma_1,\sigma_2,\tau\rangle$

$\cong\mathbb{Z}_3\times\mathbb{Z}_3\rtimes\mathbb{Z}_2$ Note $\tau\sigma_i\tau=\sigma_i^2$ $\sigma_i^3=1$ $\tau^2=1$

$\mathbb{Q}(\sqrt{2})\cap\mathbb{Q}(\sqrt[3]{2},\sqrt[3]{3},\alpha)=\mathbb{Q}$ (otherwise $\mathbb{Q}(\sqrt{2})\leftrightarrow H<Gal(\mathbb{Q}(\sqrt[3]{2},\sqrt[3]{3},\alpha))$ an subgroup of order 9, so $H=\langle\sigma_1,\sigma_2\rangle \Rightarrow \mathbb{Q}(\sqrt{2})=\mathbb{Q}(\sqrt[3]{2},\sqrt[3]{3},\alpha)^H=\mathbb{Q}(\alpha)$ ↯)

So $Gal\ \mathbb{Q}(\sqrt{2},\sqrt[3]{2},\sqrt[3]{3},\alpha)\cong\mathbb{Z}_2\times(\mathbb{Z}_3\times\mathbb{Z}_3\rtimes\mathbb{Z}_2)$

m) $x^n-t=\prod(x-\alpha^i\sqrt[n]{t})$ $\alpha=\cos\frac{2\pi}{n}+i\sin\frac{2\pi}{n}$ $t\in\mathbb{C}$ $\Rightarrow \mathbb{K}=\mathbb{C}(\sqrt[n]{t}) \Rightarrow [\mathbb{K}:\mathbb{C}(t)]=n$

$Gal(\mathbb{K}:\mathbb{C}(t))\cong\mathbb{Z}/n\mathbb{Z}$ generated by $\phi:\alpha^i\sqrt[n]{t}\to\alpha^{i+1}\sqrt[n]{t}$

n) $(x^4-t)=(x^2-\sqrt{t})(x^2+\sqrt{t})=(x+i\sqrt[4]{t})(x-i\sqrt[4]{t})(x+\sqrt[4]{t})(x-\sqrt[4]{t})$ $\mathbb{K}=\mathbb{R}(i,\sqrt[4]{t})$

$[\mathbb{K}:\mathbb{R}(i)]$ Galois $\leftrightarrow$ $\phi:\sqrt[4]{t}\to i\sqrt[4]{t}$ $\tau:\mathbb{K}\to\mathbb{K}$ $\tau(z)=\bar{z}$

$(\tau\phi\tau)(\sqrt[4]{t})=-i\sqrt[4]{t}=\phi^3(\sqrt[4]{t})$ so $\phi^4=1$ $\tau^2=1$ $\tau\phi\tau=\phi^3 \Rightarrow Gal\cong D_4$.